THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 09

Full Syllabus (Chapters 1–8) | Class 9 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(b)** $(4,\ 7)$	$\left(\frac{2+(6)}{2}, \frac{4+(10)}{2}\right)$
2	**(b)** 10 units	$\sqrt{36+64}=10$
3	**(d)** Quadrant II	signs of (x, y) are $(-, +)$
4	**(d)** 6	read off the term $6x^2$
5	**(a)** $6n-5$	common difference 6, so $a_n = 6n-5$
6	**(a)** $\frac{1}{3}$	standard method ⇒ $\frac{1}{3}$
7	**(b)** terminating	denominator 40 is of the form 2^a5^b
8	**(b)** subtracting a number from itself $(a-a=0)$	Brāhmasphuṭasiddhānta (628 CE)
9	**(a)** $(7x+2y)^2$	identify $a^2+2ab+b^2=(a+b)^2$
10	**(d)** Incorrect; the term $2ab$ is missing	$(a+b)^2=a^2+2ab+b^2$
11	**(d)** 70°	opposite angles of a cyclic quadrilateral add to 180°
12	**(b)** 24 cm	half-chord $=\sqrt{13^2-5^2}=12$, so chord $=2\times12=24$
13	**(d)** 42°	angle at the centre is double the angle on the circle
14	**(a)** 84 cm^2	area $=bh$
15	**(b)** 38.5 cm^2	$\pi r^2=\frac{22}{7}\times3.5^2=38.5$
16	**(d)** 22 cm	$\frac{60}{360}\times2\pi r=22$
17	**(c)** $\frac{2}{5}$	8 favourable out of 20 equally likely balls
18	**(d)** 600	expected number $=P\times$ population
19	**(b)** 55	$1+2+\cdots+10$
20	**(d)** 38	substitute $n=7$

SECTION — B (Very Short Answer) — Answers — 8 × 2 = 16

Q.	Expected Answer / Value Points	Marks
21	$AB=\sqrt{(5-(-3))^2+(3-(-3))^2}=\sqrt{64+36}=\sqrt{100}=\mathbf{10}$ units	1+1
22	$a_n=4n+(3)$; $a_{15}=4(15)+(3)=\mathbf{63}$	1+1
23	20 has prime factors only 2 and/or 5 (2^a5^b) ⇒ terminating	1+1
24	$(7x+y)^2$	2
25	Half-chord 20; $r=\sqrt{20^2+21^2}=\sqrt{841}=29$ cm.	1+1
26	$s=21$; area $=\sqrt{21\times14\times6\times1}=\sqrt{1764}=42$ cm^2.	1+1
27	11 letters. (i) B occurs twice ⇒ $\frac{2}{11}$. (ii) Vowels O, A, I, I ⇒ $\frac{4}{11}$.	1+1
28	$t_1=-1,\ t_2=2,\ t_3=5,\ t_4=8,\ t_5=11$.	1+1

SECTION — C (Short Answer) — Answers — 8 × 3 = 24

Q.	Expected Answer / Value Points	Marks
29	Right angle at O. Hypotenuse $AB=\sqrt{36+64}=10$ units (1½). Area $=\frac{1}{2}\times6\times8=24$ sq. units (1½).	1½+1½
30	$y=-2x$ (each y is -2 times x). Slope -2 ⇒ the line falls; y decreases by 2 for each unit increase in x (linear decay).	1+1+1
31	$\frac{x}{3}+\frac{x}{2}=\frac{5x}{6}=\frac{16}{3}\Rightarrow x=\frac{32}{5}$.	2+1
32	$(3a)^3-(2b)^3=(3a-2b)(9a^2+6ab+4b^2)$.	1+2
33	Circumference $=176$ cm (1). 1.76 km $=176000$ cm (1). Revolutions $=\frac{176000}{176}=1000$ (1).	1+1+1
34	(i) $\frac{24}{40}=\frac{3}{5}$ (ii) $\frac{16}{40}=\frac{2}{5}$ (and $\frac{3}{5}+\frac{2}{5}=1$).	1+1+1
35	$\angle BOC=2\times50^\circ=100^\circ$ (1). $OB=OC\Rightarrow\angle OBC=\angle OCB=\frac{180^\circ-100^\circ}{2}=40^\circ$ (2).	1+2

	OR: $\angle A + \angle D = 180^\circ$ (co-interior) and $\angle A + \angle C = 180^\circ$ (cyclic) ⇒ $\angle C = \angle D$ (2). Base angles equal ⇒ $AD = BC$ (1).	2+1
36	$S_{25} = \frac{25\times 26}{2} = 325$ marbles.	2+1
	OR: $S_{40} - S_{11} = \frac{40\times 41}{2} - \frac{11\times 12}{2} = 820 - 66 = 754$.	2+1

SECTION — D (Long Answer) — Answers

4 × 5 = 20

Q.	Expected Answer / Value Points	Marks
37	(i) For $a < b$ the average $\frac{a+b}{2}$ is rational and lies between them (3). (ii) $\frac{1}{2}\left(-\frac{3}{12} + \frac{5}{12}\right) = \frac{1}{12}$; it lies to the right of 0, one-twelfth of a unit from the origin (2).	3+2
	OR: (i) No, π is irrational. (ii) $1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} = \frac{105-35+21-15}{105} = \frac{76}{105}$; times $4 = \frac{304}{105} \approx 2.895$ — a rough value, improving with more terms. (iii) Lambert, 1761.	1+2+2
38	$CA = CD$ ⇒ $\angle ACE = 2\angle ADC$ (exterior angle) (2). $CB = CD$ ⇒ $\angle BCE = 2\angle BDC$ (2). Subtract: $\angle ACB = \angle ACE - \angle BCE = 2(\angle ADC - \angle BDC) = 2\angle ADB$ (1).	2+2+1
	OR: (i) 10 m rope: $\sqrt{169 - 25} = 12$ m; 24 m rope: $\sqrt{169 - 144} = 5$ m (2). (ii) $12 + 5 = 17$ m (1). (iii) $12 - 5 = 7$ m (1). (iv) 24 m rope is longer and nearer to the centre (1).	2+1+1+1
39	(i) $s = 88$ m (1). (ii) $\sqrt{63 \times 49 \times 36 \times 28} = 1764$ m^2 (2). (iii) $25^2 + 60^2 = 65^2$ and $39^2 + 52^2 = 65^2$, so both triangles are right-angled; areas $750 + 1014 = 1764$ ✓ (1). (iv) $1764 \times 20 = ₹35{,}280$ (1).	1+2+1+1
	OR: (i) $\frac{360}{8} = 45^\circ$ (1). (ii) $\frac{45}{360} \times 2\pi \times 14 = 11$ cm (1). (iii) $11 + 14 + 14 = 39$ cm (1). (iv) $\frac{45}{360} \times \frac{22}{7} \times 196 = 77$ cm^2 (2).	1+1+1+2
40	(i) 1000, 2000, 4000, 8000 (1). (ii) $500 \cdot 2^n$ (1). (iii) $500 \times 64 = 32000$ (1). (iv) $500 \cdot 2^n > 10000 \Rightarrow 2^n > 20 \Rightarrow n = 5$ (16,000) (2).	1+1+1+2
	OR: (i) AP with $a = 1$, last term $2n - 1$: $S = \frac{n}{2}(1 + 2n - 1) = n^2$ (2). (ii) 25 terms ⇒ $25^2 = 625$ (1). (iii) $1 + 3 + \cdots + 23 = 12^2 = 144$ (2).	2+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTER